

Key Terminology

Wafer

- A thin, round slice of silicon
- Used as the base to build computer chips
- One wafer holds many chips

Die (plural: dies)

- A single chip on a wafer
- Each die becomes one usable chip if it passes testing

Die-Level Testing (Wafer Probe Test)

- Electrical testing done before chips are separated
- Checks if each die works correctly

Yield

- The percentage of dies that work correctly
- Higher yield means lower manufacturing cost

Wafer Map

- A 2D picture of a wafer
- Shows which dies passed or failed testing

Pixel (in Wafer Maps)

- Represents one die
- Stores test results, not colors or brightness

Defect Pattern

- A visible shape formed by failing dies
- Often points to a manufacturing problem
- Examples: center failures, edge rings, scratches

Process Sensors

- Devices inside manufacturing machines
- Measure conditions like temperature, pressure, and power

Process Run / Production Instance

- A single execution of a manufacturing process, often corresponding to the processing of one wafer or lot under a specific set of tool conditions.

Introduction

- **Wafer map data**, which provides a spatial view of defects across a wafer after fabrication and testing
- **Process sensor data**, which captures physical and electrical conditions during wafer fabrication

While both data sources relate to the same manufacturing process, they capture fundamentally different aspects of it. Wafer maps describe *where* defects occur, whereas sensor data helps explain *why* those defects may have occurred.

Die-Level Electrical Testing

Before a wafer is cut into individual chips, it undergoes **die-level electrical testing**, commonly referred to as **wafer probe testing**.

During this stage:

- A probe card makes physical contact with each die on the wafer
- Electrical characteristics such as continuity, leakage current, and voltage thresholds are measured
- Each die is classified as passing, failing, or untested

The results of this testing stage provide the raw information used to construct wafer maps.

What Is a Wafer Map?

A **wafer map** is a two-dimensional grid representation that preserves the physical layout of dies on a wafer.

- Each grid cell corresponds to a single die
- The spatial arrangement reflects the physical location of dies on the wafer
- The circular shape of the wafer naturally emerges from the grid structure

How Wafer Map Pixels Are Generated

In wafer maps, a **pixel does not represent light intensity**, as in conventional digital images.

Instead:

- Each pixel encodes the **electrical test outcome of a single die**
- Pixel values are discrete and symbolic

Typical encoding:

- `0` → No die or untested location
- `1` → Die passed electrical test
- `2` → Die failed electrical test

Thus, each pixel corresponds directly to a physical die on the wafer.

Spatial Defect Patterns

Because dies are arranged spatially across the wafer surface, defects often form recognizable geometric patterns:

- **Center defects** — commonly associated with lithography focus or exposure issues
- **Edge-ring defects** — linked to temperature gradients or etching non-uniformity
- **Scratch defects** — caused by mechanical handling or tool contact
- **Random defects** — typically resulting from particle contamination

These spatial regularities make wafer maps well suited for image-based machine learning models such as convolutional neural networks (CNNs).

Process Sensors in Semiconductor Manufacturing

Unlike wafer maps, which are generated after fabrication, **process sensors operate during fabrication**.

Sensors are embedded within manufacturing tools such as etchers, deposition chambers, and lithography systems, and they monitor:

- Temperature
- Pressure and vacuum levels
- Gas flow rates
- Plasma density
- Radio-frequency (RF) power
- Chemical concentrations
- Tool health and calibration indicators

These sensors provide continuous insight into the conditions under which wafers are manufactured.

Example: Wafer Map Insight (Where Defects Occur)

What we observe

- Many failed dies form a ring near the wafer edge
- The pattern is consistent across multiple wafers

What this tells us

- The problem is spatial, not random
- Defects are linked to wafer position

Conclusion

- The issue likely comes from a process that affects wafer edges
- Examples: uneven heating or etching near the boundary

References

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